Mitigating Algorithmic Bias in Classification

Impact, Techniques, and Implementation

COM772

Emerging and Advancing Topics in AI

Alison McDowell

B00999060

Bias In AI

- Algorithmic bias is the systematic error in machine-learning models that leads to unequal or unfair outcomes for different social groups.
- Why it Matters:
  - Social Cost:
    - Reinforce inequality and harm affected groups as will be discussed in later slides.
  - Financial Cost:
    - Under the EU AI Act, breaching the ban on discriminatory AI practices can result in fines of up to €35 million or 7% of global annual turnover [1].
  - Reputational Cost:
    - 86% of surveyed organisations believe customers favour brands that transparently apply ethical AI practices [1].

Sources of Bias

- Data Collection Bias: Historical Imbalance
  - Historical inequalities, such as prioritising English-language internet content and commercially profitable markets [1].
  - Example: Six leading ImageNet models were 10% less accurate on images from low-income households, with even larger errors on non-Western images [1].
- Data Collection Bias: Sampling Bias
  - Sampling bias data comes from a narrow or limited source, such as a single hospital or region, reducing model generalisability [2].
  - Example: Medical imaging models trained on one institution's equipment and protocols may fail when deployed elsewhere [2].
- Annotation and Measurement Bias:
  - Human labelling errors, sensor inaccuracies, annotation mistakes or inconsistencies are correlated with protected attributes [1].
  - Example: Gender recognition systems misclassified dark-skinned women 34.7% of the time vs 0.8% for light-skinned men [1].

Sources of Bias

- Feature / Proxy Variable Bias:
  - When a feature substitutes for a protected attribute in a harmful way, such as using healthcare spending as a proxy for health status [1].
  - Example: Black patients received far fewer recommendations for additional care because historical spending was lower [1].
- Model Training Bias:
  - Hyperparameters (*e.g.,* class weights, regularisation) shifting the decision boundary in a way that can produce worse outcomes for certain groups [1].
  - Can boost overall accuracy but at the cost of a higher false-negative rate for vulnerable groups [1].
- Metric Selection Bias:
  - Choosing evaluation metrics that ignore fairness (*e.g.,* optimising only for accuracy rather than equalised odds or demographic parity) [1].

Sources of Bias

- Deployment and Feedback Loop Bias:
  - Once deployed, model decisions influence new data, causing even small initial biases to compound over time [1].
  - Example: Predictive policing system (PredPol) began almost exclusively directing police to neighbourhoods with previously recorded arrests, amplifying bias [1].

Detecting Bias: Processes

- Systematically checking a dataset for known types of bias before training a model [2]:
  - Exclusion bias: Are certain groups missing?
  - Selection bias: Were participants chosen in a way that favours one group?
  - Recall bias: Is the data affected by how it was collected or remembered?
  - Observer/annotation bias: Did the humans labelling data introduce subjectivity?
  - Prejudice bias: Does the dataset reflect existing social stereotypes?
- Perform unsupervised analysis *e.g.,* Principal Component Analysis and hierarchical clustering to identify patterns and highlight data skewness [2].
- Evaluate statistical results by subgroup, such as gender or age categories [2].
  - Visualization of algorithm output with methods such as class activation heatmaps [2].
- Conduct an independent code review [2].
  - Assumptions, feature engineering choices, and hyperparameter tuning can be coded and cause bias.

Detecting Bias: Frameworks

- The Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool [2].
  - A 14-question framework assessing the risk of bias in the study.
  - Rating of high, low or unclear.
- The Prediction Model Risk of Bias ASsessment Tool (PROBAST) [2].
  - Evaluates the risk of bias in four potential categories: participants, predictors, outcome, and analysis.
  - Risk of Bias is either high, low or unclear.
  - Current version is not fully aligned with how machine learning models are built.
- PROBAST-AI [3].
  - Currently in development.
  - Focuses on multivariable prediction model studies.

Detecting Bias: Fairness Metrics

- Demographic Parity (Statistical Parity) [9]:
  - All groups receive positive predictions at the same rate, regardless of their true outcomes.
- Equalized Odds [9]:
  - Both the true positive rate and false positive rate are equal across all groups.
- Equality of Opportunity [9]:
  - Only true positive rate is required to be equal across all groups.
- Predictive Parity [9]:
  - The probability that a positive prediction is correct, Positive Predictive Value, is equal across all groups.
- Overall Predictive Parity [9]:
  - The probability that a negative prediction is correct, Negative Predictive Value, is equal across all groups.
- Disparate Impact [35]:
  - The ratio of positive prediction rates between privileged and unprivileged groups. Guided by the U.S. 80% rule, where values below 0.8 indicate potential discrimination.

Detecting Bias: Fairness Metrics

Scores are probability-like model outputs, typically between 0 and 1, indicating how confident the model is that a person belongs to the positive class.

- Calibrated Algorithm [9]:
  - Individuals from different groups with the same score have the same likelihood of belonging to the positive class.
  - The score does not need to match the actual probability; it only needs to be consistent across groups.
- Well Calibrated Algorithm [9]:
  - The score reflects the true probability.
- Balance for the Positive Class [9]:
  - Individuals in the positive class receive the same average score across different groups.
- Balance for the Negative Class [9]:
  - Individuals in the negative class receive equal average scores across different groups.

Detecting Bias: Fairness Metrics Trade-Offs [35]

No single fairness metric satisfies all fairness goals. Choosing one often sacrifices another, especially in high-stakes domains like criminal justice.

- Demographic Parity *vs.* Equal Opportunity:
  - Statistical Parity aims for equal positive outcomes across groups.
  - Equal Opportunity ensures fairness for qualified individuals (true positives).
  - Optimizing Demographic Parity may unintentionally favour higher-risk individuals, violating Equal Opportunity.
- Equal Opportunity *vs.* Predictive Equality:
  - Equal Opportunity equalizes true-positive rates.
  - Predictive Equality focuses on equal false-positive rates.
  - Improving one can worsen the other, raising true positives may increase false positives.
- Disparate Impact *vs.* Equal Opportunity:
  - Disparate Impact ensures proportional positive outcomes across groups.
  - Equal Opportunity prioritizes fairness for those who qualify.
  - Balancing both may disadvantage highly qualified or clearly unqualified individuals.

Detecting Bias: Aequitas

- Open-source toolkit for auditing ML models for group fairness [10].
- Provides decision tree to guide fairness metric selection [10].
- Available as a Python library, command-line tool, and web interface [10].
- Generates group level metrics [10].

Fairness Metric [9]	Aequitas Group Metric [10]	Formulas [11]
Demographic Parity: Equal rate of predicted positives across groups.	**Predicted Prevalence (PP):** Fraction of group predicted positive.	PP is used directly.
Equality of Opportunity: Equal True Positive Rate (TPR) across groups.	**False Negative Rate (FNR):** Fraction of actual positives incorrectly predicted as negative.	TPR = 1 - FNR
Equalized Odds: Equal TPR and FPR across groups.	**FNR** **False Positive Rate (FPR):** Fraction of actual negatives incorrectly predicted as positive.	TPR = 1 - FNR FPR is used directly.
Overall Predictive Parity: Equal PPV across groups.	**False Discovery Rate (FDR):** Fraction of predicted positives that are incorrect (false positives).	Positive Predictive Value (PPV) = 1 - FDR

Bias Mitigations: Pre-processing

Adjusts the dataset before training to reduce bias via rebalancing datasets and data augmentation.

- Reweighting:
  - Adjusts sample weights before training so that under-represented or disadvantaged groups contribute more to model learning [5].
- Disparate Impact Remover:
  - Modifies sensitive features or correlated attributes to reduce their influence on without fully removing them so protected attributes become less predictive [5].
- Under-Sampling:
  - Reduces samples from majority groups to balance class distribution [8].
- Over-Sampling:
  - Synthetic data is generated for under-represented groups to rebalance the dataset [6].
  - Juwara *et al.* Found it improved performance in datasets which has low to medium bias [6].

Bias Mitigations: In-processing

Modifies the learning algorithm or training procedure during training to enforce fairness constraints (*e.g.*, Adversarial debiasing, fairness-aware loss functions).

- Exponentiated Gradient Reduction (ExGR):
  - Iteratively updates model parameters and weights to satisfy fairness constraints optimizing a fairness-aware objective [5].
- Meta Fair Classifier:
  - Accepts fairness metrics as constraints and trains a secondary model to adjust predictions of the primary model based on sensitive attributes [5].
- Transfer Learning:
  - Fine-tunes a baseline model trained on a majority group using minority group data so the model better represents under-represented patterns [4].
  - Gu *et al.* found this improved predictive performance across ethnic and racial groups when modelling COVID-19 mortality [4].

Bias Mitigations: Post-processing

Adjusts predictions after the model is trained to correct biased outcomes. Threshold adjustment, fairness constraints applied post-hoc.

- Reject option classification:
  - Cases near the decision boundary are left unclassified, then reassigned favourably for disadvantaged groups and unfavourably for privileged groups [7].
  - 5/8 uniform bias reduction, 2/8 mixed bias reduction, 1/8 no bias reduction [7].
- Calibration:
  - Recalibrating model outputs across subgroups to ensure predicted probabilities align with the true probabilities of outcomes [7].
  - 4/8 uniform bias reduction, 2/8 mixed bias reduction, 2/8 no bias reduction [7].
- Threshold Adjustment:
  - Modifying decision thresholds across subgroups to improve fairness by aligning error rates [7].
  - 8/9 uniform bias reduction, 1/9 no bias reduction [7].

Bias Toolkits: Fairlearn & AI Fairness 360 (AIF360) [25]

- There are more than six open-source fairness toolkits available.
- The most popular toolkits are Fairlearn and AIF360.
- Both support bias detection and mitigation.

	Fairlearn	AIF360
Open source	Yes	Yes
Organisation	Microsoft	IBM
Language Support	Python Package	R & Python Package
Scope of Use	Pre-processing, in-processing & post-processing.	Pre-processing, in-processing & post-processing.
Bias Detection Metrics	17	71
Mitigation Algorithms	4	9
Ease of Use	Easy to use with clear guidance.	More complex and steeper learning curve.

Bias Toolkits: Fairlearn & AI Fairness 360 (AIF360)

<table>
<tr><th>Fairness Metrics</th><th>Fairlearn [26, 27]</th><th>AIF360 [28, 29]</th></tr>
<tr><td>Demographic Parity</td><td>demographic_parity_difference()
demographic_parity_ratio()</td><td>metrics.statistical_parity_difference()</td></tr>
<tr><td>Equalized Odds</td><td>equalized_odds_difference()
equalized_odds_ratio()</td><td>false_positive_rate_difference() +
true_positive_rate_difference()</td></tr>
<tr><td>Equality of Opportunity</td><td>equal_opportunity_difference()
equal_opportunity_ratio()</td><td>metrics.equal_opportunity_difference()</td></tr>
<tr><td>Predictive Parity</td><td>Use sklearn.metrics.precision_score
with metric_frame.difference()</td><td>positive_predictive_value()</td></tr>
<tr><td>Overall Predictive Parity</td><td>NPV is not built-in but there are
workarounds.</td><td>positive_predictive_value() +
negative_predictive_value()</td></tr>
<tr><td>Calibrated Algorithm</td><td colspan="2">Not supported but can be coded manually.</td></tr>
<tr><td>Well Calibrated Algorithm</td><td colspan="2">Manual (binning + group comparison)</td></tr>
<tr><td>Balance for the Positive Class</td><td rowspan="2">No built-in function .
Filtering positive or negative samples
then use mean_prediction</td><td rowspan="2">No built-in function. Use scores, labels and
protected_attributes then compute it
manually.</td></tr>
<tr><td>Balance for the Negative Class</td></tr>
<tr><td>Disparate Impact (Ratio)</td><td>Manually calculate with metric_frame</td><td>metrics.disparate_impact()</td></tr>
</table>

Bias Toolkits: Fairlearn & AI Fairness 360 (AIF360)

	Fairlearn [30, 31]	AIF360 [32]
Pre-processing	• Correlation Remover • Prototype Representation Learner	• Disparate Impact Remover • Learning Fair Representations • Optimized Preprocessing • Reweighing

In-processing	• Reduction Algorithms
	    • Exponentiated Gradient
	    • Grid Search
	• Adversarial Mitigation
		• Adversarial Debiasing
		• ART Classifier
		• Gerry Fair Classifier
		• Meta Fair Classifier
		• Prejudice Remover
		• Exponentiated Gradient reduction
		• Grid search Reduction

Post-processing	• Threshold Optimizer	• Calibrated Equalized Odds • Equalized odds • Reject Option Classification • Deterministic Reranking

Ethical Issues

- Residual Bias [12]:
  - Hofmann highlights that residual or undetected biases may persist even after applying mitigation techniques.
  - It cannot be assumed that bias mitigation fully removes bias from the data.
- Individual Interpretation [13]:
  - There is no single agreed approach to ensuring fairness.
  - Everyone will have their own interpretation of fairness; it is inherently a normative decision.
  - The decision can advantage or disadvantage certain groups over others.
- Algorithmic Formalism [14]:
  - A formal rule-based abstraction tends to ignore real world complexity.
  - It can create a misleading sense of neutrality and objectivity, potentially reinforcing existing inequalities.
  - Moral oversimplification can hide deeper structural injustices that remain unaddressed.

Social Issues

- Lack of Intersectionality [15, 16]:
  - Traditional fairness approaches often focus on a single attribute (*e.g.,* race or gender).
  - Intersectional identities are ignored (*e.g.,* race + gender + socioeconomic status) which can lead to worse outcomes for certain subgroups.
- Erosion of Trust [17, 18]:
  - Public trust may decline if fairness definitions, mitigation methods, or results appear arbitrary or inconsistent.
  - Transparency is essential, as perceived unfairness can undermine legitimacy and acceptance of AI systems.
- Structural Blindness [19]:
  - Focusing on bias mitigation within data can miss the broader ways technology and humans conspire to reproduce discrimination.
  - Fails to address the structural inequalities that shape who benefits and who is marginalised.

Legal Issues

- Fairness *vs*. Legal Definitions [13]:
  - Fairness definitions may conflict with legal definitions of non-discrimination.
  - Mitigation strategies which are "fair" may conflict with legal frameworks on non-discrimination, privacy or data protection.
  - Legal restrictions on using protected attributes may limit how bias can be mitigated.
- Accountability Challenges [12]:
  - When an AI system "debiased" via bias mitigation techniques causes harm, it can be unclear who is legally responsible.
- Enforcement of Standards [21]:
  - The EU AI Act mandates measures are undertaken to detect, prevent and mitigate bias.
  - Most Member States lack dedicated national authorities to monitor AI compliance.
  - Bias mitigation processes and algorithms are proprietary making legal oversight and auditing difficult.

Professional Issues

- Trade-offs [22]:
  - No bias mitigation technique achieves the ideal balance between fairness and model performance.
  - Engineers must make trade-offs and decide whether to prioritise fairness or accuracy.
- Responsibility Ambiguity [23]:
  - There is uncertainty over who is responsible for detecting and mitigating bias, leaving AI professionals unsure of their level of responsibility.
  - This lack of clarity can lead to a diffusion of responsibility with no party feeling fully responsible for ensuring fairness.
- Cross-Domain Coordination [24]:
  - Guidance is needed beyond technical fixes, requiring collaboration across legal, ethical, organisational, and technical domains.
  - Such coordination is complex and resource-intensive for practitioners.

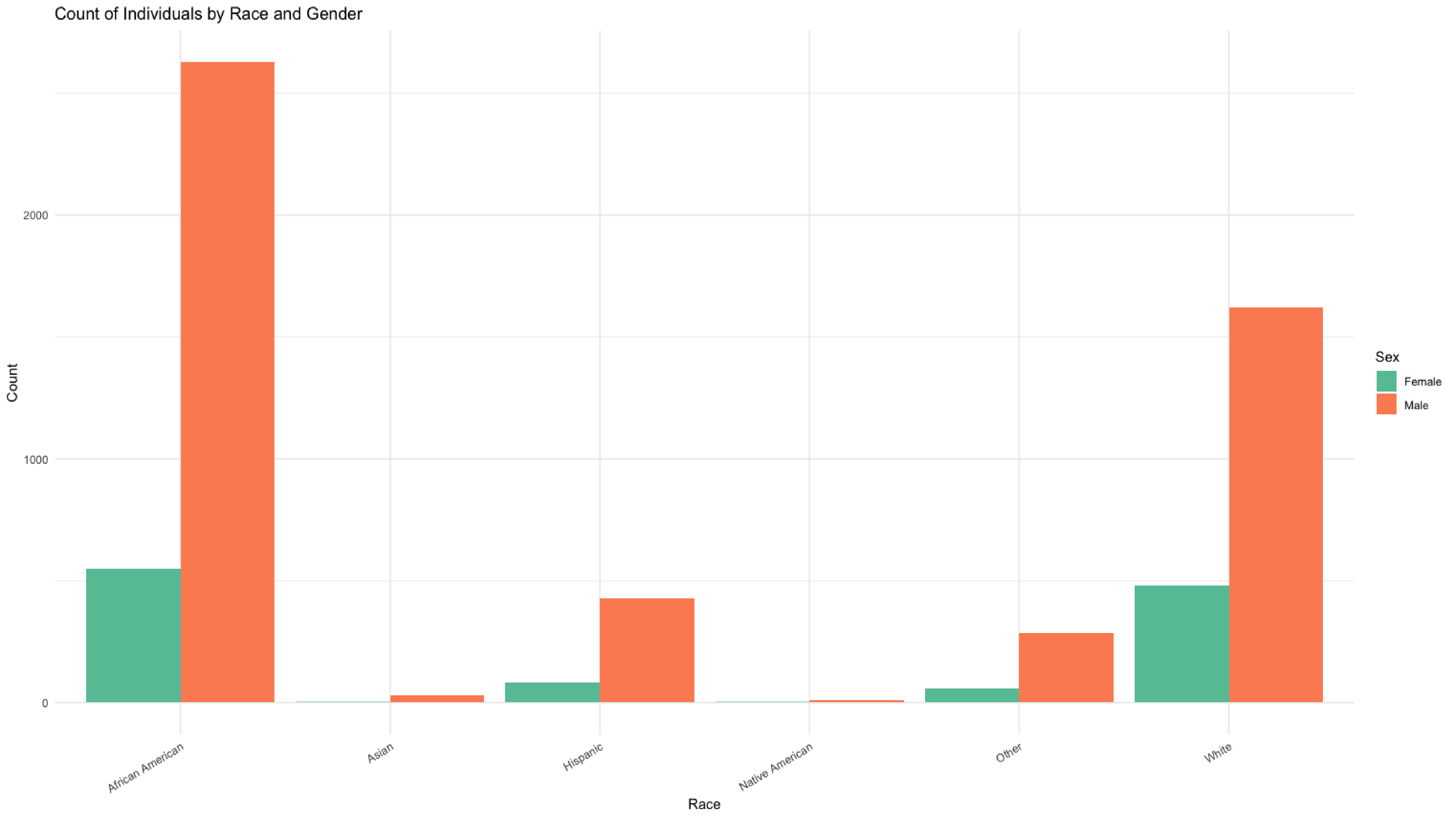
COMPAS Dataset [33, 34]

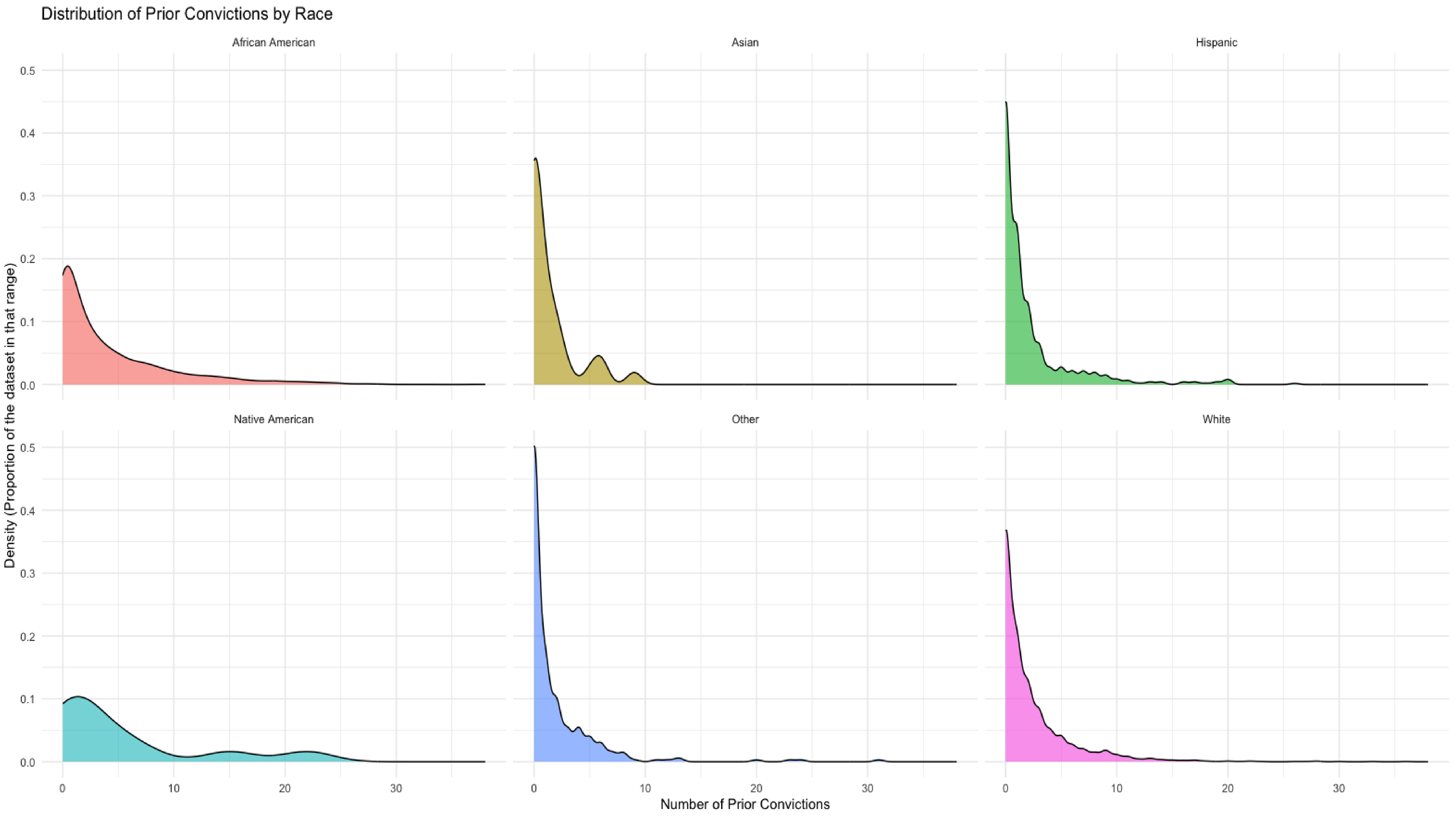
- COMPAS (Correctional Offender Management Profiling for Alternative Sanctions).
- A widely deployed commercial risk-scoring algorithm used by US courts and parole boards to predict an individual's probability of recidivism.
- Includes 137 features about individuals and their criminal history.
- Although race is not used directly as an input, other variables correlated with race may act as proxies.
- A 2016 ProPublica investigation found that COMPAS falsely flagged Black defendants as high-risk more often than white defendants, despite similar re-offence patterns. This raised concerns about:
  - Racial bias
  - Lack of transparency (algorithm is proprietary)
  - Consequences of automated decision-making in sentencing fairness

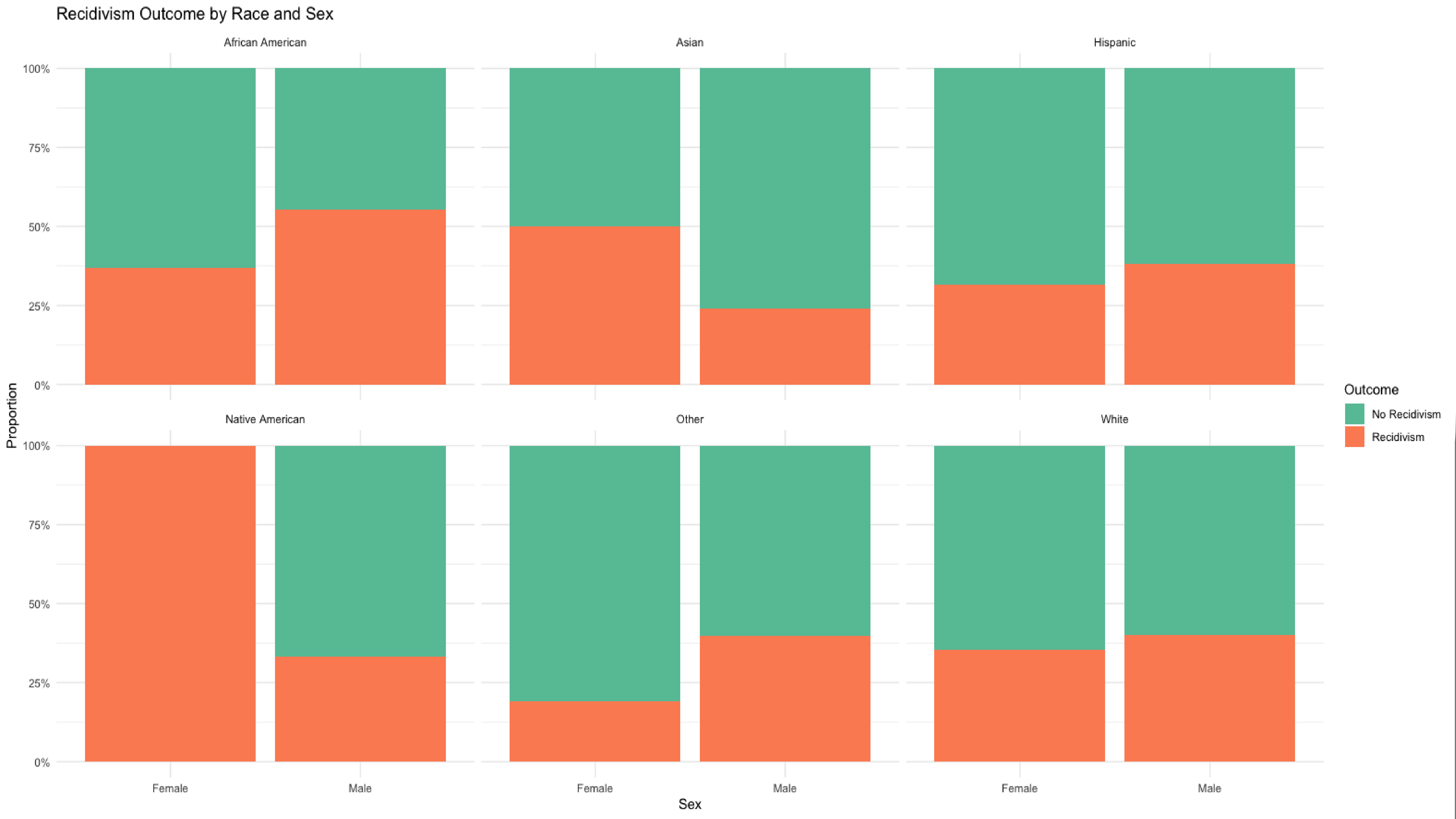
COMPAS Dataset

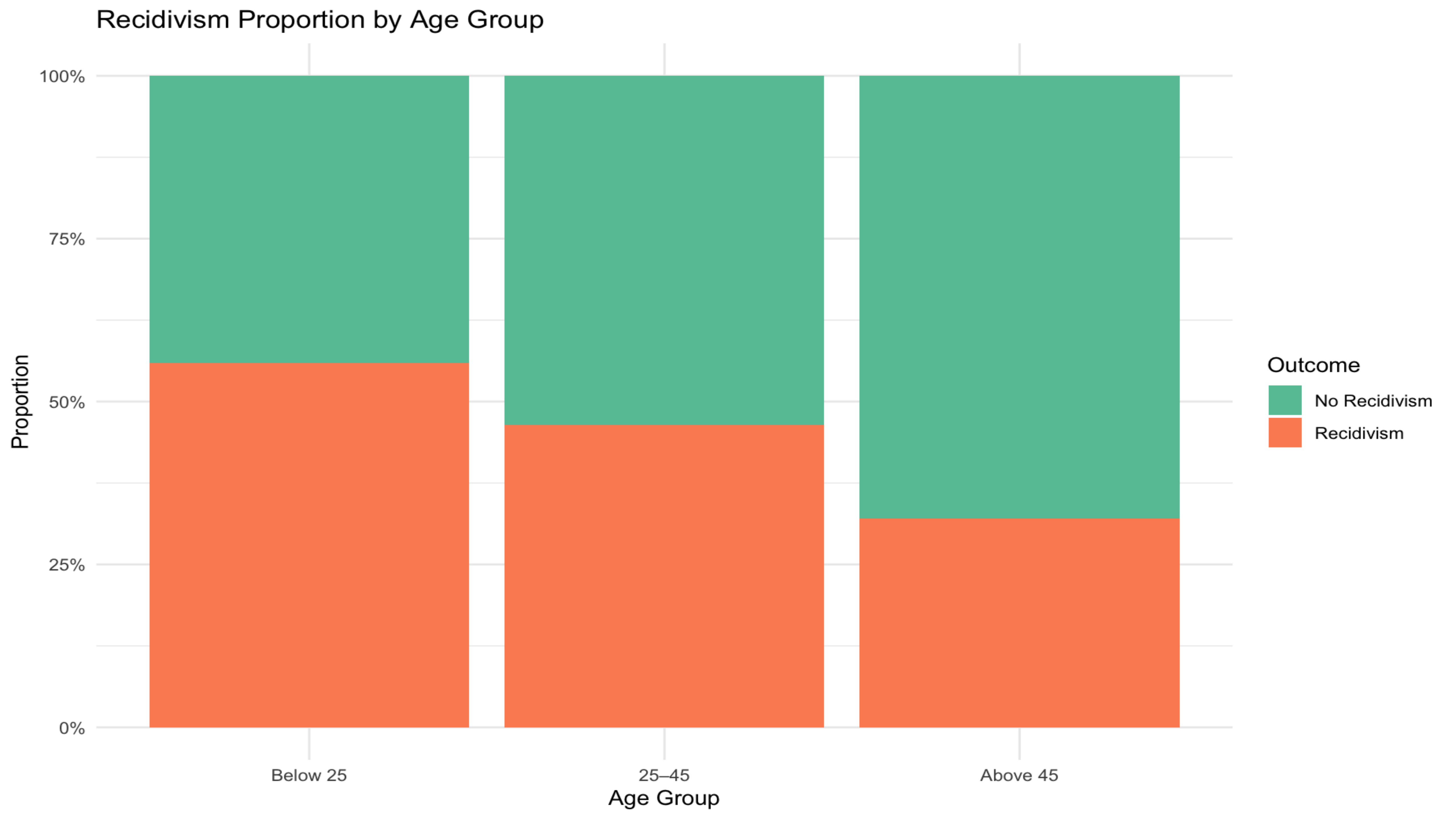
Feature	Description
Two_yr_Recidivism	Target variable. Rearrest within 2 years (1 = yes, 0 = no).
Number_of_Priors	Count of previous convictions.
score_factor	Binary indicator (Medium/High = 1, Low = 0).
Age_Above_FourtyFive	Binary indicator (1 = age > 45, 0 = otherwise).
Age_Below_TwentyFive	Binary indicator (1 = age < 25, 0 = otherwise).
African_American	Binary indicator (1 = African American, 0 = not).
Asian	Binary indicator (1 = Asian, 0 = not).
Hispanic	Binary indicator (1 = Hispanic/Latino, 0 = not).
Native_American	Binary indicator (1 = Native American, 0 = not).
Other	Binary indicator (1= not covered by other categories, 0 = not).
Female	Binary indicator for sex (1 = female, 0 = male).
Misdemeanor	Binary indicator (1 = misdemeanor charge, 0 = felony).

- Propublica_data_for_fairml.csv
- 6172 records.
- Cleaned and simplified version of the original ProPublica COMPAS dataset.

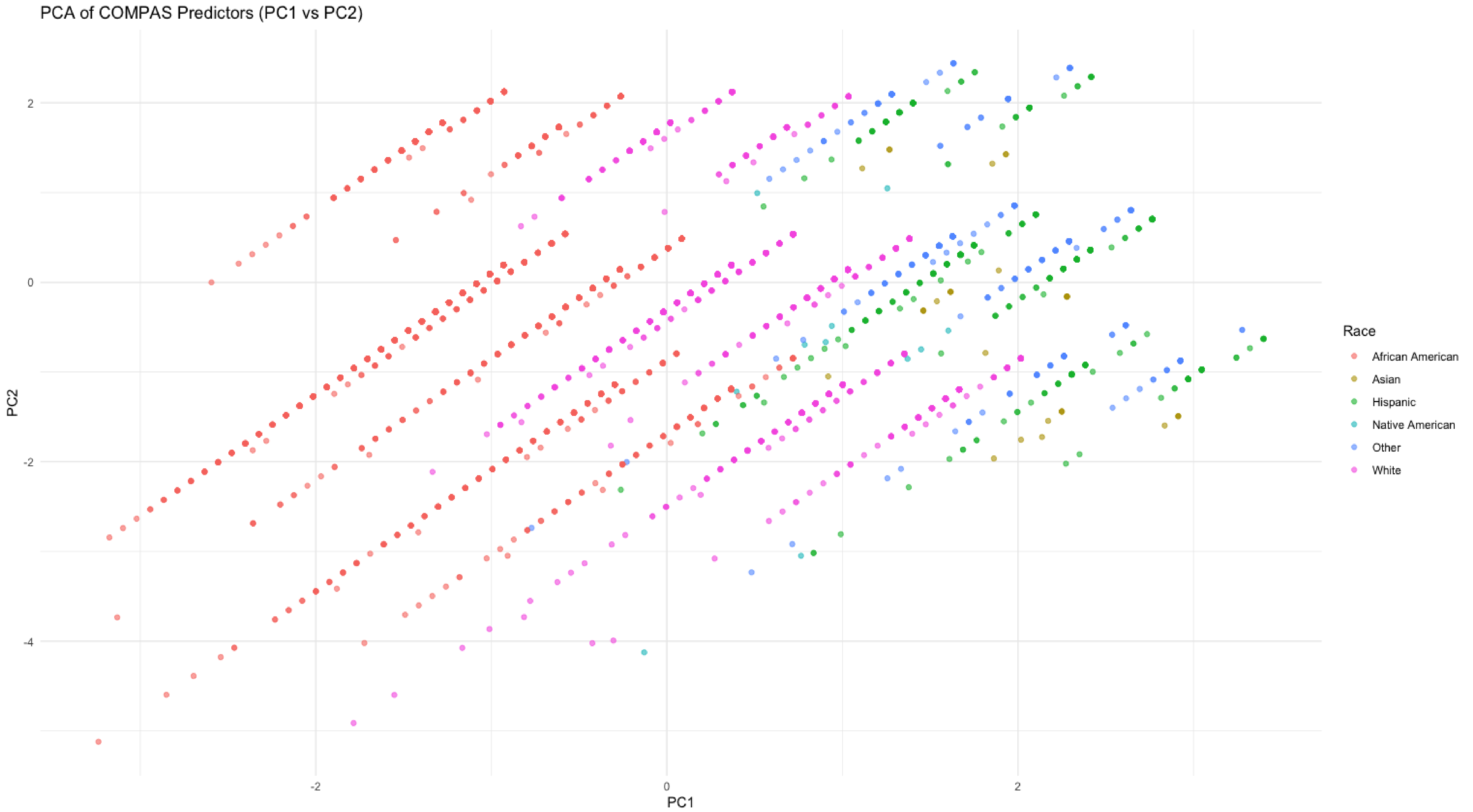

Count of Individuals by Race and Gender
Count
2000
1000
0
African American
Asian
Hispanic
Native American
Other
White
Race
Sex
Female
Male


Distribution of Prior Convictions by Race
African American
Asian
Hispanic
Native American
Other
White
Density (Proportion of the dataset in that range)
Number of Prior Convictions
0.0
0.1
0.2
0.3
0.4
0.5
0
10
20
30


Recidivism Outcome by Race and Sex
African American
Asian
Hispanic
Native American
Other
White
100%
75%
50%
25%
0%
Proportion
Female
Male
Sex
Outcome
No Recidivism
Recidivism



Recidivism Proportion by Age Group
100%
75%
50%
25%
0%
Proportion
Below 25
25–45
Above 45
Age Group
Outcome
No Recidivism
Recidivism


PCA of COMPAS Predictors (PC1 vs PC2)
PC1
PC2
2
0
-2
-4
Race
African American
Asian
Hispanic
Native American
Other
White

COMPAS Dataset

Fairness Metric	Result	Meaning [9] [35]
Demographic Parity Difference	-0.245	Ideal: 0. African-American defendants are less likely to receive a low-risk prediction, indicating potential bias.
Equalized Odds TPR	-0.203	Ideal: 0. The system is less accurate at identifying true reoffenders in the unprivileged group.
Equalized Odds FPR	-0.212	Ideal: 0. African-American defendants are more likely to be falsely flagged as high-risk.
Equality of Opportunity	-0.203	Ideal: 0. The negative value shows unequal opportunity, meaning the tool is less fair in correctly predicting reoffending for African-American individuals.
Positive Predictive Parity Diff	Priv: 0.71 Unpriv: 0.649 Diff: -0.061	Ideal Diff: 0. High-risk predictions are less reliable for African-American individuals, indicating more incorrect harmful classifications.
Negative Predictive Parity Diff	Priv: 0.595 Unpriv: 0.65 Diff: 0.055	Ideal Diff: 0. Low-risk predictions are slightly more accurate for African-American defendants but less accurate for white defendants.
Disparate Impact (Ratio)	0.634	Ideal: 1. Below 0.8 indicates discrimination. African-Americans receive favorable predictions at 63.4% the rate of White defendants.

- Dataset: propublica_data_for_fairml.csv
- Tool: AIF360 (Python)
- Protected Group: African_American
- Privileged Group: White
- Favourable Label: 0 (No reoffence)
- Unfavourable Label: 1 (Reoffended)

COMPAS Dataset: Mitigate Bias [35]

- Farayola *et al.* (2025) found that using Disparate Impact Remover, Adversarial Learning, and Equalized Odds together (DIR+AL+EO) delivers significant fairness gains while keeping accuracy largely unchanged.
- Disparate Impact Remover (DIR):
  - A pre-processing method which adjusts feature values to reduce correlations with the protected attribute, thereby promoting fairer treatment of demographic groups.
- Adversarial Learning (AL) / Adversarial Debiasing:
  - An in-processing method trains a classifier alongside an adversarial network that attempts to predict the protected attribute, encouraging the model to avoid encoding discriminatory patterns.
- Equalized Odds Optimization (EO) :
  - A post-processing method which adjusts predicted labels after training to ensure true-positive and false-positive rates are aligned between privileged and unprivileged groups.

COMPAS Dataset

Metric	COMPAS	Baseline	DIR	AL	EO	Combination
Accuracy	0.661	0.664	0.666	0.681	0.611	0.61
Demographic Parity Difference	-0.245	-0.235	-0.258	-0.158	0.035	0.022
Equalized Odds TPR	-0.203	-0.151	-0.158	-0.082	0.032	-0.03
Equalized Odds FPR	-0.212	-0.252	-0.295	-0.153	0.097	0.14
Equality of Opportunity	-0.203	-0.151	-0.158	-0.082	0.032	-0.03
PPV Difference	-0.061	-0.028	-0.01	-0.057	-0.147	-0.18
NPV Difference	0.055	0.079	0.094	0.079	0.098	-0.017
Disparate Impact	0.634	0.674	0.65	0.777	1.049	1.029
Positive Class Balance	N/A	0.1059	0.1321	-0.0505	0.1059	-0.1017
Negative Class Balance	N/A	0.0866	0.1113	-0.0242	0.0866	-0.0696
Calibration	N/A	0.0603	0.0705	0.0605	0.0603	0.0521

References

[1] T. Maripova, “Mitigating Algorithmic Bias in Predictive Models,” *The American Journal of Engineering and Technology*, vol. 07, no. 05, pp. 192–201, May 2025, doi: 10.37547/tajet/Volume07Issue05-19.

[2] B. Koçak *et al.*, “Bias in artificial intelligence for medical imaging: fundamentals, detection, avoidance, mitigation, challenges, ethics, and prospects,” *Diagnostic and Interventional Radiology*, vol. 31, no. 2. Galenos Publishing House, pp. 75–88, Mar. 01, 2025. doi: 10.4274/dir.2024.242854.

[3] A. Navarro and C. Lourdes, *QUALITY OF MACHINE LEARNING PREDICTION MODELS IN HEALTHCARE*. Utrecht University, 2023.

[4] T. Gu *et al.*, “Mitigating bias in AI mortality predictions for minority populations: a transfer learning approach,” *BMC Medical Informatics and Decision Making*, vol. 25, no. 1, Dec. 2025, doi: 10.1186/s12911-025-02862-7.

[5] D. Gándara, H. Anahideh, M. P. Ison, and L. Picchiarini, “Inside the Black Box: Detecting and Mitigating Algorithmic Bias Across Racialized Groups in College Student-Success Prediction,” *AERA Open*, vol. 10, Jan. 2024, doi: 10.1177/23328584241258741.

References

[6] L. Juwara, A. El-Hussuna, and K. el Emam, "An evaluation of synthetic data augmentation for mitigating covariate bias in health data," *Patterns*, vol. 5, no. 4, Apr. 2024, doi: 10.1016/j.patter.2024.100946.

[7] S. Mackin, V. J. Major, R. Chunara, and R. Newton-Dame, "Post-processing methods for mitigating algorithmic bias in healthcare classification models: An extended umbrella review," *BMC Digital Health*, vol. 3, no. 1, Aug. 2025, doi: 10.1186/s44247-025-00166-4.

[8] W. Zhou, C. Liu, P. Yuan, and L. Jiang, "An Undersampling Method Approaching the Ideal Classification Boundary for Imbalance Problems," *Applied Sciences*, vol. 14, no. 13, Jul. 2024, doi: 10.3390/app14135421.

[9] P. Garg, J. Villasenor, and V. Foggo, "Fairness Metrics: A Comparative Analysis," in *Proceedings - 2020 IEEE International Conference on Big Data, Big Data 2020*, Institute of Electrical and Electronics Engineers Inc., Dec. 2020, pp. 3662–3666. doi: 10.1109/BigData50022.2020.9378025.

[10] P. Saleiro *et al.*, "Aequitas: A Bias and Fairness Audit Toolkit," *arXiv*, Apr. 2019, [Online]. Available: http://arxiv.org/abs/1811.05577

References

[11] D. Powers, "EVALUATION: FROM PRECISION, RECALL AND F-MEASURE TO ROC, INFORMEDNESS, MARKEDNESS & CORRELATION," *arXiv*, 2011.

[12] B. Hofmann, "Biases in AI: acknowledging and addressing the inevitable ethical issues," *Frontiers in Digital Health*, vol. 7, 2025, doi: 10.3389/fdgth.2025.1614105.

[13] M. Rovatsos, B. Mittelstadt, and A. Koene, "Landscape Summary: Bias in Algorithmic Decision-Making," 2019. Accessed: Dec. 03, 2025. [Online]. Available: https://assets.publishing.service.gov.uk/media/5d31c30a40f0b64a8099e21d/Landscape_Summary_-_Bias_in_Algorithmic_Decision-Making.pdf

[14] A. Tsamados *et al.*, "The ethics of algorithms: key problems and solutions," *AI and Society*, vol. 37, no. 1, pp. 215–230, Mar. 2022, doi: 10.1007/s00146-021-01154-8.

[15] K. Kobayashi and Y. Nakao, "One-vs.-One Mitigation of Intersectional Bias: A General Method to Extend Fairness-Aware Binary Classification," *arXiv*, Mar. 2025, [Online]. Available: http://arxiv.org/abs/2010.13494

[16] C. Wisdom, "Mitigating Intersectional Bias in Machine Learning A Novel Approach to Fairness in Automated Decision-Making.," *zenodo*, Nov. 2024.

References

[17] S. Nah, J. Luo, and J. Joo, "Mapping Scholarship on Algorithmic Bias: Conceptualization, Empirical Results, and Ethical Concerns," *International Journal of Communication*, vol. 18, pp. 548–569, 2024, [Online]. Available: http://ijoc.org.

[18] N. Krco, T. Laugel, V. Grari, J.-M. Loubes, and M. Detyniecki, "When mitigating bias is unfair: multiplicity and arbitrariness in algorithmic group fairness," *arXiv*, May 2024, [Online]. Available: http://arxiv.org/abs/2302.07185

[19] A. L. Hoffmann, "Where fairness fails: data, algorithms, and the limits of antidiscrimination discourse," *Information Communication and Society*, vol. 22, no. 7, pp. 900–915, Jun. 2019, doi: 10.1080/1369118X.2019.1573912.

[20] M. Pavlovic, "Expected Calibration Error (ECE): A Step-by-Step Visual Explanatoin," https://medium.com/data-science/expected-calibration-error-ece-a-step-by-step-visual-explanation-with-python-code-c3e9aa12937d. Accessed: Dec. 03, 2025. [Online]. Available: https://medium.com/data-science/expected-calibration-error-ece-a-step-by-step-visual-explanation-with-python-code-c3e9aa12937d

[21] G. F. Lendvai and G. Gosztonyi, "Algorithmic Bias as a Core Legal Dilemma in the Age of Artificial Intelligence: Conceptual Basis and the Current State of Regulation," *Laws*, vol. 14, no. 3, Jun. 2025, doi: 10.3390/laws14030041.

References

[22] Z. Chen, J. Zhang, F. Sarro, and M. Harman, "A Comprehensive Empirical Study of Bias Mitigation Methods for Machine Learning Classifiers," *arXiv*, vol. 2657, pp. 1–9, 2022, doi: 10.48550/arXiv.2207.03277.

[23] C. M. Lancaster, K. Schulenberg, C. Flathmann, N. J. McNeese, and G. Freeman, "'It's Everybody's Role to Speak Up... But Not Everyone Will': Understanding AI Professionals' Perceptions of Accountability for AI Bias Mitigation," *ACM Journal on Responsible Computing*, vol. 1, no. 1, pp. 1–30, Mar. 2024, doi: 10.1145/3632121.

[24] R. Schwartz, A. Vassilev, K. Greene, L. Perine, A. Burt, and P. Hall, "Towards a standard for identifying and managing bias in artificial intelligence," Mar. 2022. doi: 10.6028/NIST.SP.1270.

[25] H. Pandey, "Comparison of the usage of Fairness Toolkits amongst practitioners:AIF360 and Fairlearn," 2022. [Online]. Available: https://www.uclalawreview.org/injustice-ex-machina-

[26] Fairlearn, "Common fairness metrics," https://fairlearn.org/main/user_guide/assessment/common_fairness_metrics.html. Accessed: Dec. 03, 2025. [Online]. Available: https://fairlearn.org/main/user_guide/assessment/common_fairness_metrics.html

References

[27] Fairlearn, "fairlearn.metrics.MetricFrame," https://fairlearn.org/main/api_reference/generated/fairlearn.metrics.MetricFrame.html. Accessed: Dec. 03, 2025. [Online]. Available: https://fairlearn.org/main/api_reference/generated/fairlearn.metrics.MetricFrame.html

[28] aif360, "Group fairness metrics," https://aif360.readthedocs.io/en/stable/modules/sklearn.html#group-fairness-metrics. Accessed: Dec. 03, 2025. [Online]. Available: https://aif360.readthedocs.io/en/stable/modules/sklearn.html#group-fairness-metrics

[29] Hoffmansc, "metrics," https://github.com/. Accessed: Dec. 03, 2025. [Online]. Available: https://github.com/Trusted-AI/AIF360/blob/main/aif360/sklearn/metrics/metrics.py

[30] Fairlearn, "User Guide," https://fairlearn.org/v0.13/user_guide/index.html. Accessed: Dec. 03, 2025. [Online]. Available: https://fairlearn.org/v0.13/user_guide/index.html

[31] Fairlearn, "fairlearn.adversarial.AdversarialFairnessClassifier," https://fairlearn.org/main/api_reference/generated/fairlearn.adversarial.AdversarialFairnessClassifier.html. Accessed: Dec. 03, 2025. [Online]. Available: https://fairlearn.org/main/api_reference/generated/fairlearn.adversarial.AdversarialFairnessClassifier.html

References

[32] aif360, “Algorithms,” https://aif360.readthedocs.io/en/stable/modules/algorithms.html. Accessed: Dec. 03, 2025. [Online]. Available: https://aif360.readthedocs.io/en/stable/modules/algorithms.html

[33] J. Dressel and H. Farid, “The accuracy, fairness, and limits of predicting recidivism,” *Science Advances*, no. 4, Jan. 2018.

[34] D. Ofer, “COMPAS Recidivism Racial Bias,” https://www.kaggle.com/datasets/danofer/compass. Accessed: Dec. 03, 2025. [Online]. Available: https://www.kaggle.com/datasets/danofer/compass

[35] M. M. Farayola, I. Tal, T. Saber, R. Connolly, and M. Bendechache, “A fairness-focused approach to recidivism prediction: implications for accuracy, trust, and equity,” *AI and Society*, 2025, doi: 10.1007/s00146-025-02452-1.

Thank you for listening.